

PROJECT: 1

Data Cleaning & Validation Report

Dataset Size: 1,200 Records | Status: 100% Clean & Verified

1. Environment Setup & Data Preservation

- **Backup and Data Preservation:** Saved an untouched copy of the original dataset before beginning the data-cleaning process to preserve the source data and provide a reference for comparison.
- **Table Conversation:** Structured the raw data into a formal Excel Table to improve filtering, sorting, and data management.
- **Visual Styling:** Applied a professional corporate theme using dark-blue heads and alternating light-blue banded rows to improve readability and presentation.

2. Missing Values Analysis (Coupon Code)

- **Quantifying Omissions:** Identified **309 missing values** in the Coupon Code field out of **1,200 total records**.
- **Cross-Field Investigation:** The missing values were evaluated against the Oder Status field to identify potential patterns in the missing data. The 309 blank entries were distributed across multiple statuses:
 - Cancelled: 58
 - Delivered: 52
 - Shipping: 64
 - Pending: 59
 - Returned: 76
- **Distribution of Existing Data:** The remaining **891 complete records** contained three distinct coupon codes:
 - SAVE10: 286
 - FREESHIP: 313
 - WINTER15: 292
- **Remediation Strategy:** Because there was no logical basis for imputing a specific promotional code, the **309 blank entries were standardized to "NO COUPON"**. This preserved all records for analysis without introducing artificial or unsupported values.

3. Duplicate Records Audit

- **Primary Key Verification:** Conducted a comprehensive duplicate check across all **1,200 records** using the Oder ID column as the unique identifier.

- **Uniqueness Findings:** Confirmed that the dataset contains exactly **1200 unique Oder ID entries**, confirming that each record has a unique identifier.
- **Data Retention:** Because **zero duplicate entries** or redundant transactions were identified, no records were removed. This preserved the full scope of the original dataset.

4. Data Consistency Check

The dataset was reviewed for consistency across key fields, including **Order Status, Payment Method, Product, Date, Customer ID, Quantity, Unit Price, Shipping Address, Tracking Number, Item in Cart, Referral Source, Coupon Code, and Total Price.**

- Categorical fields were checked for consistent spelling and formatting.
- Dates were verified to follow the **YYYY-MM-DD** format.
- Customer ID were confirmed to follow the **C + 5-digit** format.
- Quantity values were consistently stored and ranged from **1 to 5**.
- Numeric fields, including Unit Price and Total Price, were verified as numeric.
- All **1,200 records** contained Shipping Address, Tracking Number, and Item in Cart values.
- **0 duplicate Tracking Numbers** were identified.
- Coupon Code blanks were handled separately during the missing-value analysis.
- 0 records contained a zero or negative Total Price.

Conclusion: No major data-consistency issues were identified. The dataset was considered sufficiently consistent for further analysis.

5. Outlier Check Notes

- An **IQR (Interquartile Range) analysis** was performed on the Total Price column. **Eight potential outliers** were identified and individually reviewed by comparing Quantity * Unit Price with Total Price.
- All eight records were correctly calculated and were therefore considered **valid transactions**. No records were removed or changed.

6. Final Cleaning Decisions

- After completing the data-cleaning checks, the dataset was reviewed for **missing values, duplicate records, consistency issues, validation errors, and outliers.**
- No invalid records were identified that required deletion. The **eight potential Total Price outlier** were investigated and confirmed as legitimate transactions because Quantity * Unit Price matched Total Price.
- Therefore, **no valid records were removed or modified**, and the dataset is ready for further analysis.